Krishna Khong

Sydney, Australia (Australian Citizen) · krishna@khong.dev · portfolio.khong.dev · github.com/cashewna

Final-year UNSW Computer Science student (2023–2026) building tested backend and systems software in TypeScript, Rust, Java, and SQL, alongside self-hosted Kubernetes (`k3s`), GitOps, observability, and Linux operations. Backed by software support experience and five years of customer operations.

EDUCATION

University of New South Wales (UNSW) · Bachelor of Computer Science — 2023 – 2026 · Sydney, Australia

Coursework & engineering focus: C, C++, Python, Java, TypeScript, and Rust; unit, integration, and end-to-end testing in GitHub Actions CI.

SELECTED PROJECTS

Relay Desk (`ticket-queue`) · demo.ticket-queue.khong.dev — TypeScript · Node · Fastify · PostgreSQL

- Built a server-rendered IT service-desk application with strict TypeScript, PostgreSQL SQL migrations, role-based access control, priority-computed SLA clocks, and an append-only audit log enforced at the database level.
- Modelled an explicit 5-state workflow (`new -> assigned -> in_progress -> resolved -> closed`) verified by 30 unit/integration tests against PostgreSQL and a Docker Compose HTTP acceptance suite in CI; hosted as an auto-resetting Kubernetes demo.

EPI Trainer · epi-trainer.khong.dev — React 19 · TypeScript · Spring Boot · Java 21

- Built a study console and spaced-review scheduler backed by a Spring Boot judge that compiles and runs Java solutions against EPIJudge TSV test suites, returning the first failing input, expected, and actual values.
- Verified across 96 Vitest frontend/domain tests, 33 JUnit 5 backend tests, Python data-pipeline tests, and an automated browser smoke test.

netregister — Rust · SQLite · Docker Compose

- Implemented an ARP and TCP port local network asset register in Rust with SQLite migrations and idempotent MAC-keyed upserts preserving `first_seen` history.
- Enforced target-network allow-listing before sending packets (exit code `2` on violation); verified by 5 rootless integration tests and an isolated multi-container Docker Compose ARP lab proof.

ai-eval-checker — TypeScript · Node.js (Zero Dependencies)

- Built an offline evaluation harness scoring IT ticket classification and reply drafting against a 52-ticket hand-labelled golden set across 6 classes, generating per-class precision/recall/F1, confusion matrices, and failure diagnostics.
- Verified by 60 automated unit tests and a zero-network CI reproducibility check (`env -u`).

SELF-HOSTED INFRASTRUCTURE & SECOPS

Oracle Cloud k3s Platform & GitOps (`bee-gitops` / `Infrastructure`) — NixOS · OpenTofu · Argo CD

- Provisioned an OCI Always Free Arm server (`2 OCPU / 12 GB`) with OpenTofu tenancy quota and budget guardrails; deployed single-node `k3s` reconciled via Argo CD, Traefik ingress, `cert-manager` (DNS-01 TLS), and OpenBao + External Secrets Operator.
- Built observability with VictoriaMetrics (`vmagent`, `vmalert`, Alertmanager), Grafana, blackbox, node, and kube-state exporters; operated Stalwart Mail + Bulwark JMAP webmail with OpenTofu-provisioned AWS SES (`ap-southeast-2`) outbound fallback, CrowdSec firewall bouncer, and Headscale/Tailscale mesh VPN.
- Previously operated a Proxmox VE homelab running VMs and containers with Caddy reverse proxy/TLS, self-hosted Gitea, Plex, and automated media services.

EXPERIENCE

Junior Software Developer · TFS Timeframe Systems — 2023 · Perth, Australia

- Reproduced and diagnosed client-reported software and environment defects across Windows, macOS, and Linux, isolating application bugs from OS and hardware configuration issues.
- Collaborated with developers to trace root causes, validate practical fixes, and document clear resolution steps for non-technical clients.

Sales Associate · Shahi Investments Pty Ltd — 2020 – 2025 · Perth, Australia

- Scoped customer needs, resolved enquiries and escalations end-to-end, supported inventory operations, and trained new staff on store procedures.

Service Crew · Each-A-Cup — 2011 – 2013 · Singapore

- Processed high-volume customer orders and payments accurately while maintaining stock and store operations during peak trading periods.

TECHNICAL SKILLS

Languages: TypeScript / JavaScript, Rust, Java, C / C++, Python, SQL
Systems & Web: Node.js, Fastify, Spring Boot, React, PostgreSQL, SQLite, Linux (NixOS), Windows, macOS
Infra, SecOps & Observability: Kubernetes (`k3s`), Argo CD, OpenTofu / Terraform, Docker, Proxmox VE, Traefik, Caddy, OpenBao, CrowdSec, Tailscale / Headscale, Stalwart, AWS SES, Gitea, VictoriaMetrics, Grafana, Alertmanager, Blackbox Exporter
Testing & CI: GitHub Actions CI, Vitest, JUnit 5, `node --test`, `cargo test`, deterministic fixtures